

25. The file *collegetown* contains data on 500 houses sold in Baton Rouge, LA during 2009–2013.

Variable descriptions are in the file *collegetown.def*.

a. Estimate the log-linear model $\ln(PRICE) = \beta_1 + \beta_2 SQFT + e$. Interpret the estimated model parameters. Calculate the slope and elasticity at the sample means, if necessary.

In this model, the intercept, $\beta_1 = 4.3939$, represents the predicted value of the natural logarithm of the house price when the house size is zero square feet. The slope coefficient, $\beta_2 = 0.0360$, indicates that for each additional square foot, the natural logarithm of the house price increases by approximately 0.036. This implies that, on average, the house price increases by about 3.6% for every extra square foot.

The marginal change in price with respect to square footage is calculated as:

$$\frac{d PRICE}{d SQFT} = \beta_2 \times \text{mean}(PRICE)$$

The elasticity of price with respect to square footage is calculated as:

$$Elasticity = \beta_2 \times \text{mean}(SQFT)$$

With $\text{mean}(SQFT) \approx 27.2821$ and $\text{mean}(PRICE) \approx 250.2369$, the slope in price units is approximately $0.0360 \times 250.2369 \approx 9.0085$, indicating that adding one square foot increases the house price by about \$9.01. The elasticity at the sample mean is approximately $0.0360 \times 27.2821 \approx 0.9822$, showing that a 1% increase in house size leads to roughly a 0.98% increase in price.

b. Estimate the log-log model $\ln(PRICE) = \alpha_1 + \alpha_2 \ln(SQFT) + e$. Interpret the estimated parameters. Calculate the slope and elasticity at the sample means, if necessary.

In this model, the intercept, $\alpha_1 = 2.0497$, represents the predicted value of $\ln(PRICE)$ when $\ln(SQFT)$ is zero, which corresponds to $SQFT = 1$. The slope coefficient, $\alpha_2 = 1.0248$, indicates that a 1% increase in square footage is associated with approximately a 1.02% increase in house price.

In a log-log model, the slope coefficient directly represents elasticity, which is constant across all observations. Therefore, the elasticity of price with respect to square footage is 1.0248. The slope is calculated as:

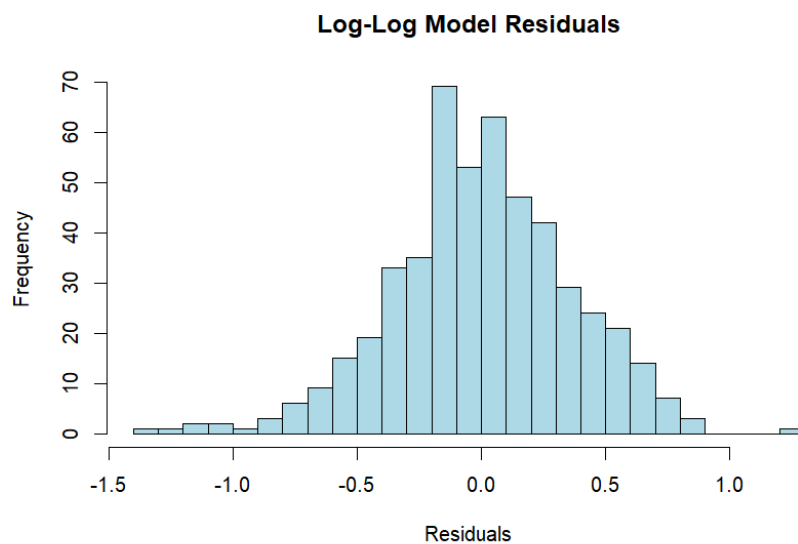
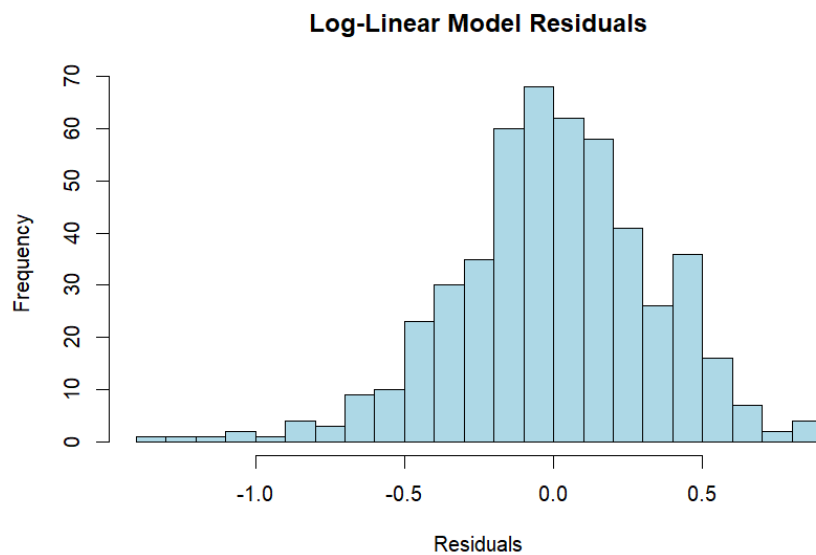
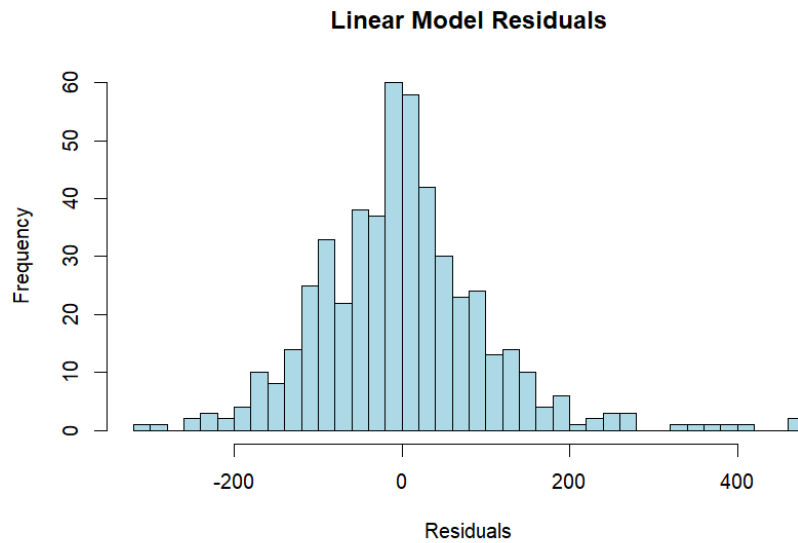
$$\frac{d PRICE}{d SQFT} = \alpha_2 \times \frac{\text{mean}(PRICE)}{\text{mean}(SQFT)}$$

Evaluated at the sample mean, this is $1.0248 \times 250.2369 / 27.2821 \approx 9.3999$, indicating that an increase of one square foot raises the house price by about \$9.40 at the average house size.

c. Compare the R^2 value from the linear model $PRICE = \delta_1 + \delta_2 SQFT + e$ to the “generalized” R^2 measure for the models in (b) and (c).

model	R^2	generalized R^2
linear model	0.6413167	0.6413167
log-linear model	0.5417259	0.6621612
log-log model	0.4738445	0.6445084

d. Construct histograms of the least squares residuals from each of the models in (a)–(c) and obtain the Jarque–Bera statistics. Based on your observations, do you consider the distributions of the residuals to be compatible with an assumption of normality?



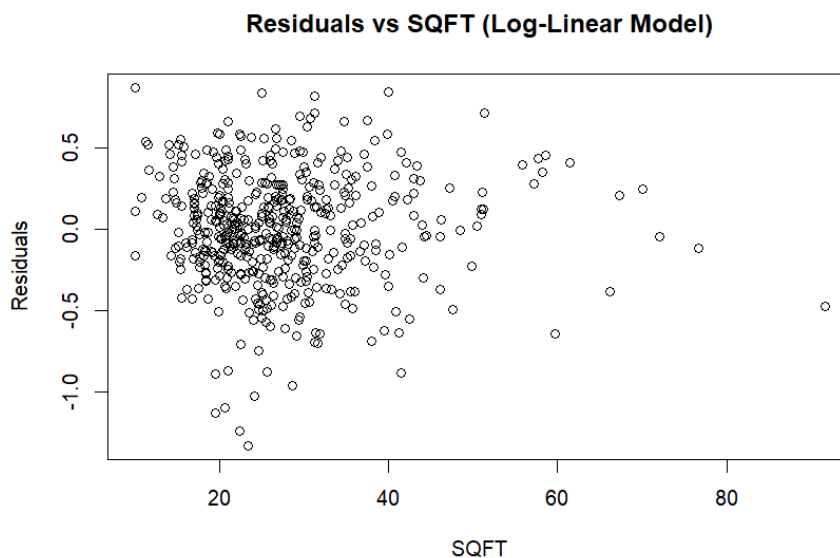
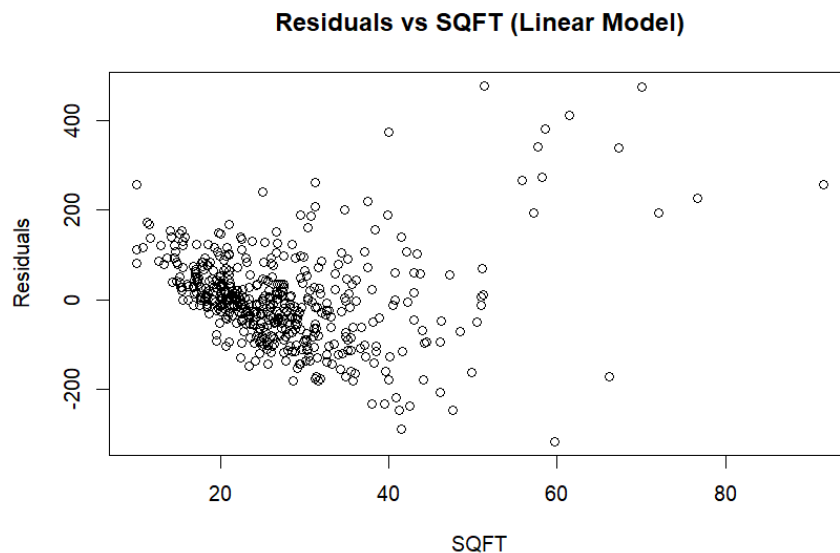
- Jarque–Bera Test

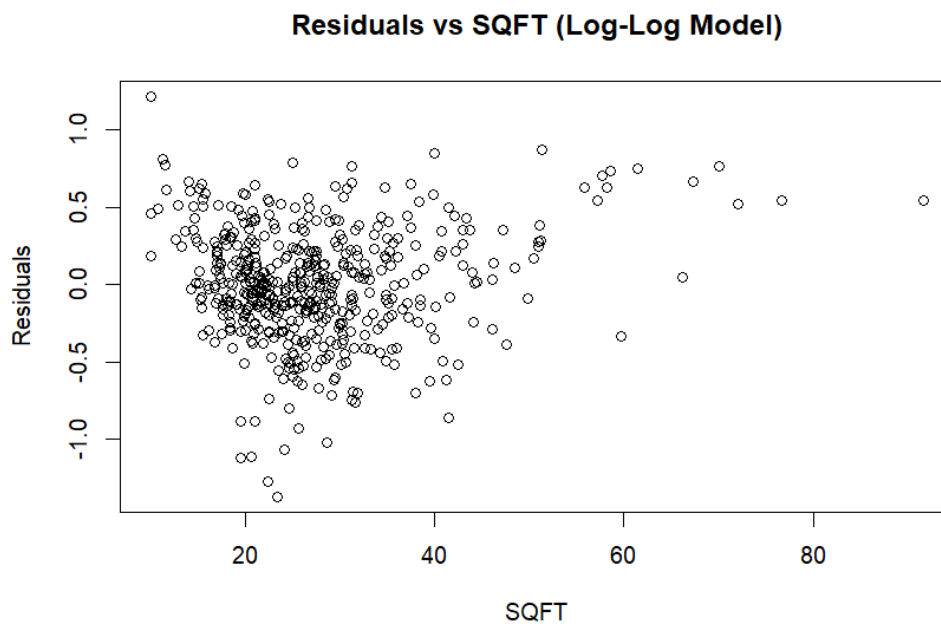
model	p-value
linear model	2.2×10^{-16}
log-linear model	1.61×10^{-6}
log-log model	0.0007933

Jarque-Bera test is usually used to test whether the data follows a Normal distribution. The null hypothesis is that the data follows a Normal distribution, and the alternative hypothesis is that the data does not follow a Normal distribution.

The p-values are all below 0.05, indicating that the residuals significantly deviate from normality. Therefore, the residuals of all three models are not consistent with the assumption of normality.

e. For each of the models in (a)–(c), plot the least squares residuals against *SQFT*. Do you observe any patterns?





Residuals from the linear model show an upward trend with SQFT, suggesting nonlinearity and increasing variance. The log-linear and log-log models reduce this pattern, with the log-log model having residuals most randomly scattered around zero.

f. For each model in (a)–(c), predict the value of a house with 2700 square feet.

model	predict value
linear model	246,445.7
log-linear model	214,233.6
log-log model	227,538.6

In the data, SQFT represents 100 square feet, and PRICE represents \$1000. Therefore, we substitute $SQFT = 2700/100 = 27$, and multiply the predicted value by 1000 to get the actual house price.

g. For each model in (a)–(c), construct a 95% prediction interval for the value of a house with 2700 square feet.

model	95% prediction interval(\$1,000)
linear model	[44.27727, 448.6341]
log-linear model	[109.7685, 418.1167]
log-log model	[111.1406, 465.8406]

h. Based on your work in this problem, discuss the choice of functional form. Which functional form would you use? Explain.

Based on the results from this problem, the choice of functional form affects both interpretability and predictive performance. The linear model has the highest traditional R^2 , but the residuals show non-normality and increasing variance with house size, suggesting it may not handle skewness well. The log-

linear and log-log models reduce these issues, with the log-linear model achieving the highest generalized R^2 , indicating better predictive accuracy for actual house prices. Considering residual patterns, elasticity interpretation, and prediction intervals, the **log-linear model** is the most appropriate functional form for modeling house prices in this dataset.